Table of Contents

ACKNOWLEDGEMENT……...…...3

ABSTRACT…….……....4

LIST OF FIGURES…….....5

1 . INTRODUCTION……6

1.1 Introduction……....……7

1.2 Literature Survey……..……..…..8

1.2.1 Existing System.…….....9

1.2.2 Proposed System.……..……...……10

2. PROBLEM STATEMENT…….…….....11

2.1 Problem Statement……..……..…..12

2.2 Scope……..…12

2.3 Software development approach.…….……...13

3. SYSTEM DESIGN……14

3.1 System Architecture……..…15

3.2 Modules and their Description ……....16

3.3 Data Flow Diagram.…….…..…17

3.3.1 DFD Level 0:…….…….....17

3.3.2 DFD Level 1:……..……18

3.3.3 DFD Level 2:……....…….……..……..19

3.4 Flow Chart…….……..……..20

3.5 Class Diagram.……..…….21

4. IMPLEMENTATION AND EXPERIMENTAL SETUP…….……22

4.1 Tools……....23

4.2 Latest Versions……...……....23

5. EXPERIMENTAL RESULTS AND IT'S ANALYSIS……..…...…24

5.1 Create Account……...…….....…25

5.2 Manage Existing Account……..…….25

5.3 Exit……....28

6. CONCLUSION……...…29

Conclusion…...……...……..30

7. REFERENCES……...…31

7.1 Web References…….……32

7.2 Book Reference…….….32

7.3 Paper Reference…….32

ACKNOWLEDGEMENT

We would like to express our gratitude to words respected guide Prof. M. K. Hasabe (HOD) for his constant encouragement and valuable guidance during the completion of this project. We are very thankful to Hon. Principal, Dr. D.V.Ghewade for their valuable co-operation & guidance during this project work.

We take this opportunity to thank the entire staff member for their co-operation and their helpfulness during this project work. Last but not the least assistant offered by various friends and colleagues related directly or indirectly to this work is also gratefully acknowledged.

Ms. Aliya Salim Mulla
Ms. Bhagyashri Pandurang Kanade
Ms. Bushra Riyazahamad Nadaf
Ms. Pushpa Dattatray Kamble

ABSTRACT

Bank Management System (BMS) is a digital platform designed to streamline and automate the daily operations of a bank, providing secure, efficient, and effective management of financial transactions, customer accounts, and banking services. The system allows banks to manage a broad array of activities, such as account creation, deposits, withdrawals, loans, transaction processing, and customer relationship management. By implementing a BMS, banks can reduce operational costs, minimize errors, improve customer service, and ensure compliance with regulatory requirement. This system typically includes modules for customer information management, account transactions, loan processing, and reporting, offering real-time data access to both customers and administrators. Advanced security measures, such as encryption and multi-factor authentication, ensure data protection and fraud prevention. Additionally, a BMS facilitates regulatory compliance by providing audit trails and supporting data retention policies.

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	System Architecture	15
3.2	Modules and their Description	16
3.3.1	DFD level 0	17
3.3.2	DFD level 1	18
3.3.3	DFD level 2	19
3.3.4	Flow Chart	20
3.3.5	Class Diagram	21
5.1	Create Account	25
5.2.1	Deposit	25
5.2.2	Withdraw	26
5.2.3	Display Account Details	26
5.2.4	Take Loan	27
5.2.5	Repay Loan	27
5.2.6	Display Loan Details	28
5.3	Exit	28

Chapter 1

Introduction

1. INTRODUCTION

1.1 INTRODUCTION:

The phone application Bank Management System is based on a concept of recording customer's account details. Here the user can perform all the tasks like creating an account, deposit amount, withdraw amount, check balance, view all account holders' detail, close an account, and modify an account. There is login system for this project. All the main features for banking system are set in this project.

The project Bank Management system includes registration of clients, storing their details talking about the features of the Bank Management System, a user can create an account by providing the name of the account holder, account number, select amount type whether its Saving account or Current account and providing an initial amount. Then the user can also deposit and withdraw money just by providing his/her account, then the system displays his/her profile and entering an amount. For certain purpose, he/she can also check for the balance inquiry which displays the account holder's name with account number type and amount. He/she can also check for all the account holder's list. Another feature is that the user can also close their account by providing their account number and he/she can modify their account detail and type if they want to.

The Bank management system deals with the concept of making a complete record of a customer's account. The system utilizes key concepts from Object-Oriented Programming and allows users to manage their bank accounts in an interactive way through the following operations:

- Creating a new bank account
- Depositing money into an account
- Withdrawing money from an account
- Checking the balance of an account
- Displaying the details of all account holders
- Closing an account

1.2 LITERATURE SURVEY:

1.2.1 Existing System:

An existing bank management system typically involves manual processes, often relying on paper records and physical work, which can be slow, inefficient, and prone to errors. These systems often lack robust security measures for customer data and bank information, and can struggle to handle large volumes of transactions efficiently.

Manual Processes:

- Slow and Inefficient:

Every task, from opening an account to processing transactions, is often handled manually, leading to delays and reduced efficiency.

- High Workload:

Manual data entry, file management, and paper-based record keeping require significant manpower.

- Increased Error Rates:

Manual processes are susceptible to human error, leading to discrepancies and inaccuracies in financial records.

- Limited Security:

Paper records and manual systems lack robust security measures, making them vulnerable to data breaches and fraud.

- Scalability Issues:

As the bank's customer base grows and transaction volumes increase, manual systems struggle to keep up.

- Lack of Automation:

Existing systems often lack automated features, making it difficult to streamline processes and reduce manual effort.

- Time-Consuming:

Manual data entry, updating records, and generating reports are time-consuming processes that can impact efficiency.

1.2.2 Proposed System:

Who can use the system?

A Bank Management System is used by various users including customers who can check balances and make transactions, employees who handle daily operations like deposits and loans, administrators who manage the system and generate reports, system admins who ensure security, and auditors who monitor for compliance. Each user group has specific access and responsibilities.

Functionality of Admin:

In a Bank Management System, the admin plays a central role in overseeing and managing the overall system. They are responsible for handling user accounts, managing customer data, monitoring transactions, and ensuring security protocols are followed. The admin also has the authority to approve or deny customer requests, manage employee access to the system, generate reports for audits, and over see system performance. They ensure smooth operations by maintaining data integrity, setting user permissions, and responding to any issues or anomalies that may arise within the system.

Functionality of Member / Owner of Bank Account:

The functionality of a **Member** or **Owner** of a bank account depends on the account type (individual, joint, business) and the bank, but generally includes depositing and withdrawing funds, managing transactions, and accessing account information through online or mobile banking. Account holders can set security preferences, avoid overdrafts, and ensure the account remains in good standing. In joint accounts, multiple people share rights and responsibilities. For business accounts, owners or authorized members may manage business transactions and apply for loans. They may also close the account, provide tax-related information, and designate beneficiaries. Special accounts, like trust or minor accounts, may involve a guardian or trustee. Overall, account holders have the rights to manage and access funds, but they are responsible for maintaining the account and following the bank's terms.

Chapter 2

Problem Statement

2. PROBLEM STATEMENT

2.1 Problem statement:

A problem statement for a bank management system highlights the inefficiencies and challenges in current banking operations, aiming to improve processes, enhance security, and provide better customer service through a modern, integrated system. Taking deposits from their customers and issuing current or checking accounts and savings accounts to individuals and business. Extending loans to individuals and business, Cashing cheque, etc.

2.2 Scope:

- Creating New Accounts- The application can be used to create two different types of accounts by the customers, which are Savings Account and Current Account. It helps save the hustle for the customer to visit the bank physically and create/use these accounts.
- Depositing Money- As the world is moving towards the limited use of paper currency, depositing or transferring money from one bank to the other will become as easy as clicking a few buttons using this application.
- Withdrawing Money- Requests can be sent through the application to ask for money transfer as well.
- Account Holder List- This is a feature for the admin. The admin can view the list of all the account holders.
- Balance Enquiry- The customer can check their balance via this application.

2.3 Software development approach:

A robust software development approach for a Bank Management System (BMS) should prioritize security, scalability, and compliance, while also leveraging modern technologies and best practices like agile methodologies, DevOps, and cloud-native architectures. This ensures the system is not only functional but also adaptable to the evolving banking landscape and regulatory requirements.

Here's a more detailed look at key aspects of such an approach:

1. Security by Design:

Secure SDLC:

Integrate security considerations throughout the entire development lifecycle, from planning to deployment, to prevent vulnerabilities from being introduced.

Zero-Trust Security:

Adopt a zero-trust approach, requiring authentication and authorization for every access request, even within the network.

Encryption:

Utilize robust encryption techniques for data at rest and in transit, protecting sensitive customer information.

Multi-Factor Authentication:

Implement multi-factor authentication for user access and sensitive transactions.

AI-Driven Fraud Detection:

Leverage AI and machine learning to detect and prevent fraudulent activities.

2. Agile Development & DevOps:

Agile Methodologies:

Use agile methodologies like Scrum for iterative development, allowing for flexibility and responsiveness to evolving requirements.

DevOps Automation:

Implement DevOps practices, such as continuous integration and continuous delivery (CI/CD) pipelines, to automate testing, deployment, and maintenance processes.

Close Collaboration:

Foster close collaboration between development and operations teams throughout the entire software lifecycle.

3. Modern Technology Stack:

Programming Languages:

Utilize languages like Java, Python, C#, and Swift, known for their performance, scalability, and extensive frameworks.

Frameworks & Libraries:

Leverage popular frameworks like Spring for Java, Django for Python, and .NET for C# to streamline development.

Cloud Platforms:

Embrace cloud platforms like AWS, Azure, and Google Cloud for scalability, flexibility, and cost-effectiveness.

Chapter 3

System Design

3. SYSTEM DESIGN

A Bank Management System (BMS) is a critical application used by banks to handle various services, such as managing customer accounts, processing transactions, managing loans, and overseeing employees. An effective database design ensures smooth operations, secure data handling, and efficient query execution.

3.1 System Architecture:

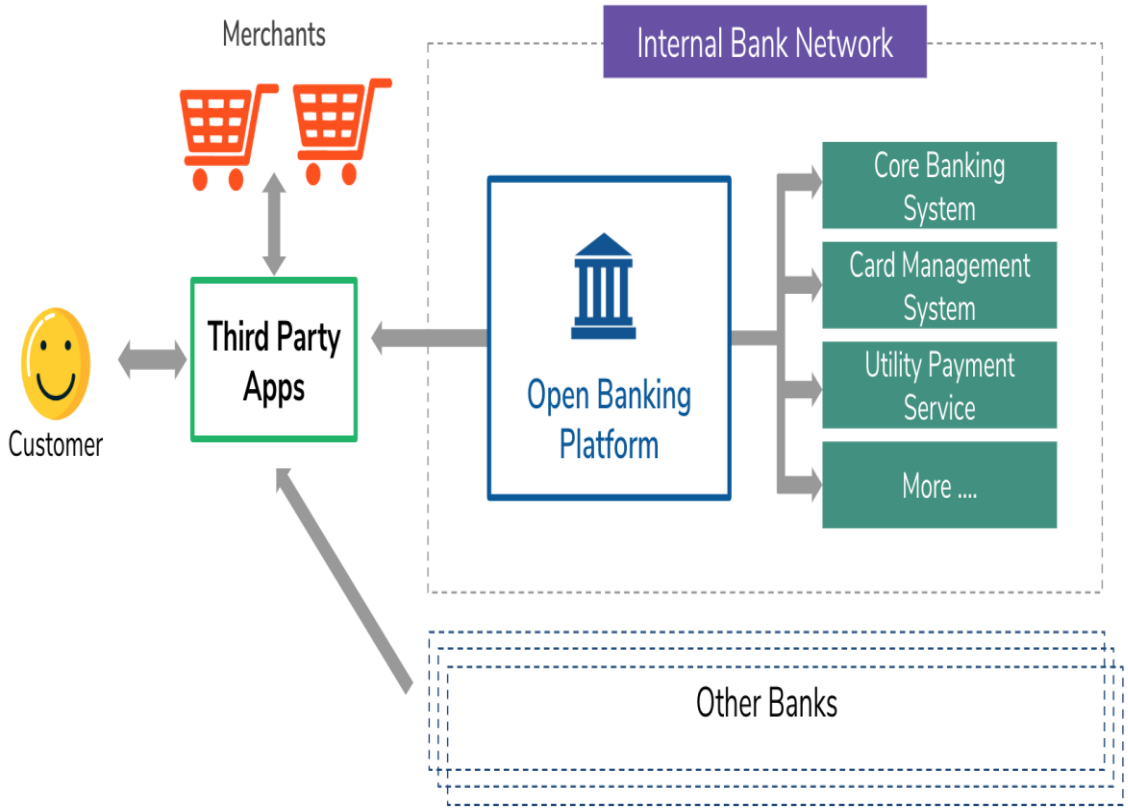


Fig: 3.1 System Architecture Diagram

3.2 Modules and their Description:

Modules	Description
Create file	Selecting this creates a new file for the users by accepting input such as account number , name and amount
Open account	Opens a new account for the users by accepting input such as account number, name and minimum balance
search	Enables to search for the details of the given account number displays only one account detail at a time
Deposit	Provide option to deposit amount from the given account number
withdraw	Provide option to withdraw amount from the given account
Cancel account	Cancels the selected account from the bank

3.3 Data Flow Diagram

3.3.1 DFD LEVEL 0:

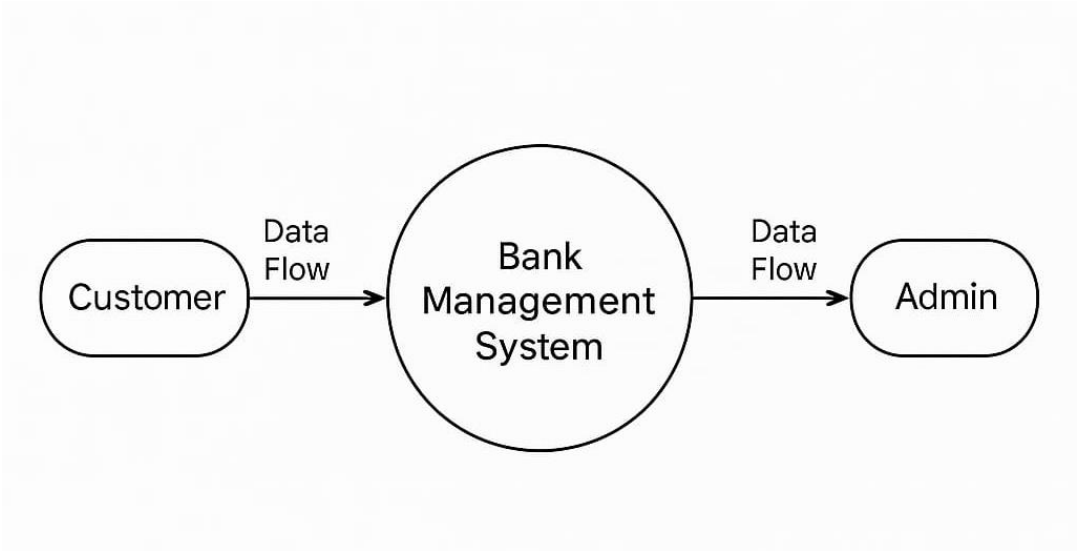


Fig: 3.3.1 DFD Level 0

3.3.2 DFD LEVEL 1:

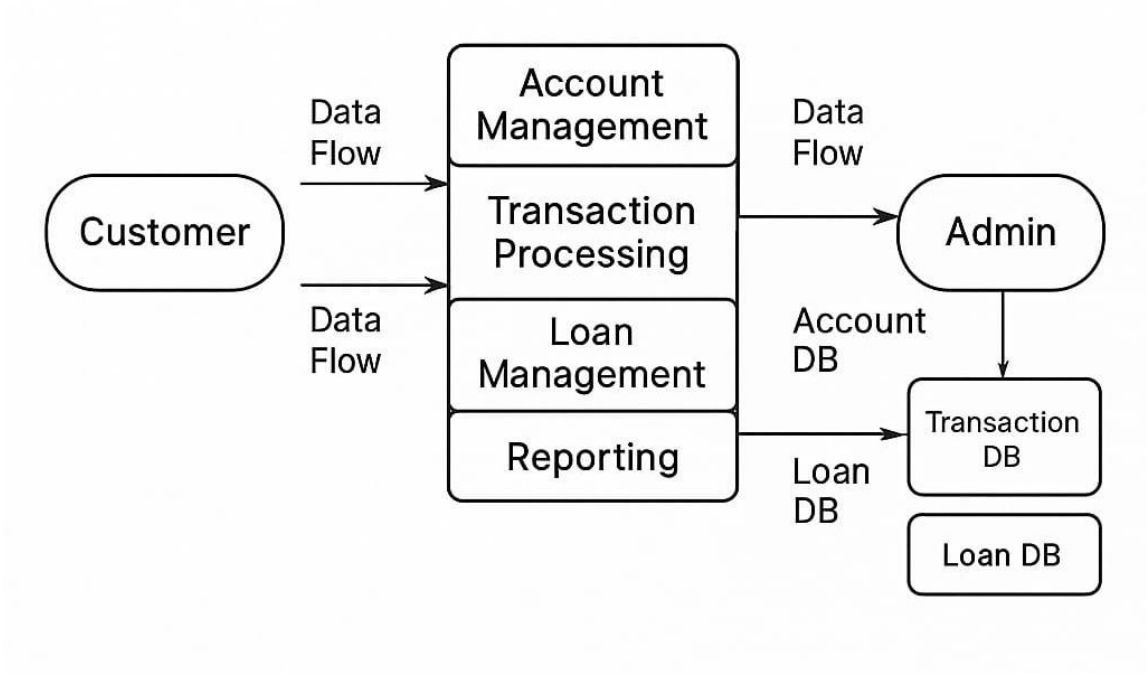


Fig: 3.3.2 DFD Level 1

3.3.3 DFD LEVEL 2:

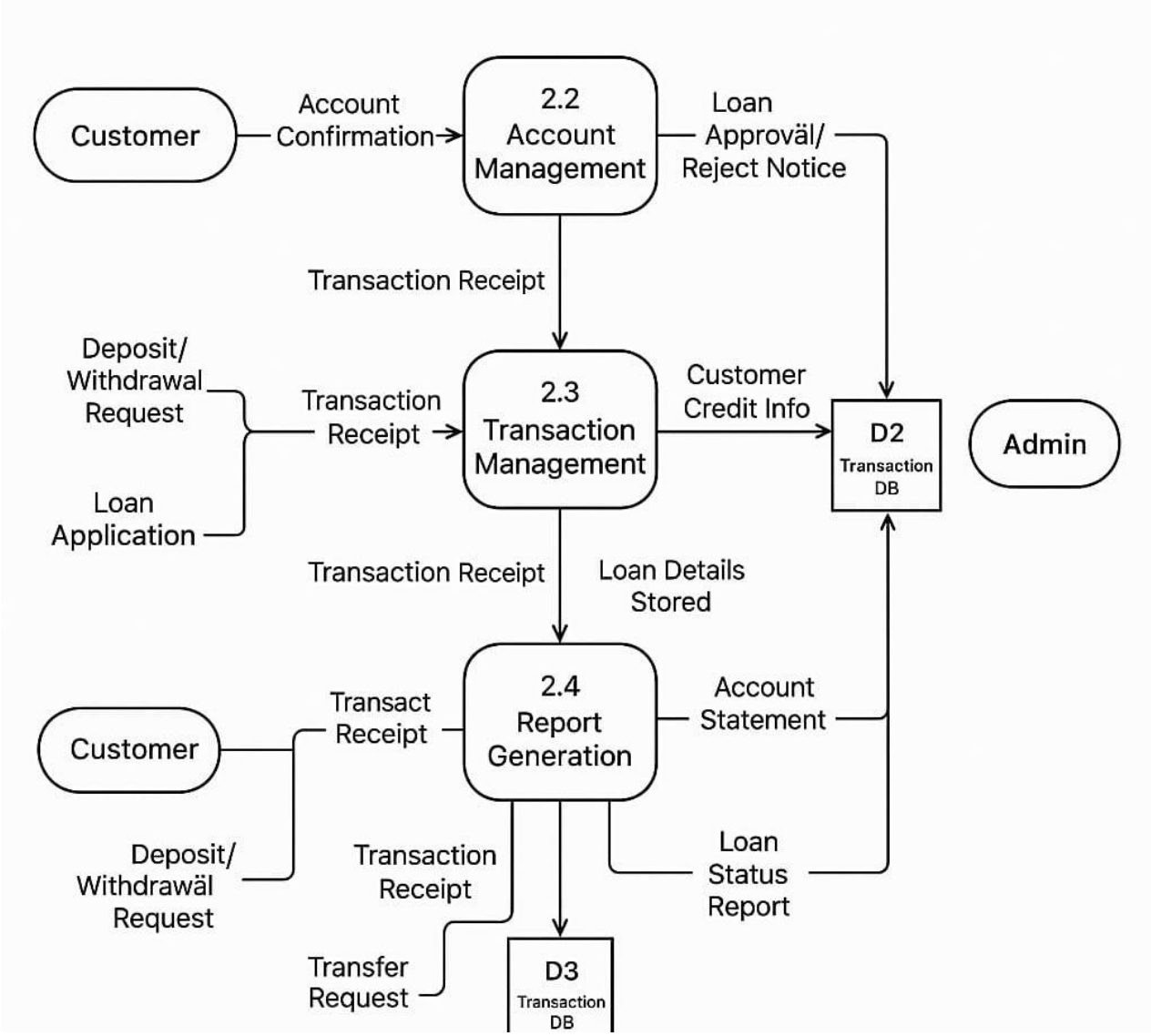


Fig: 3.3.3 DFD Level 2

3.4 Flow Chart

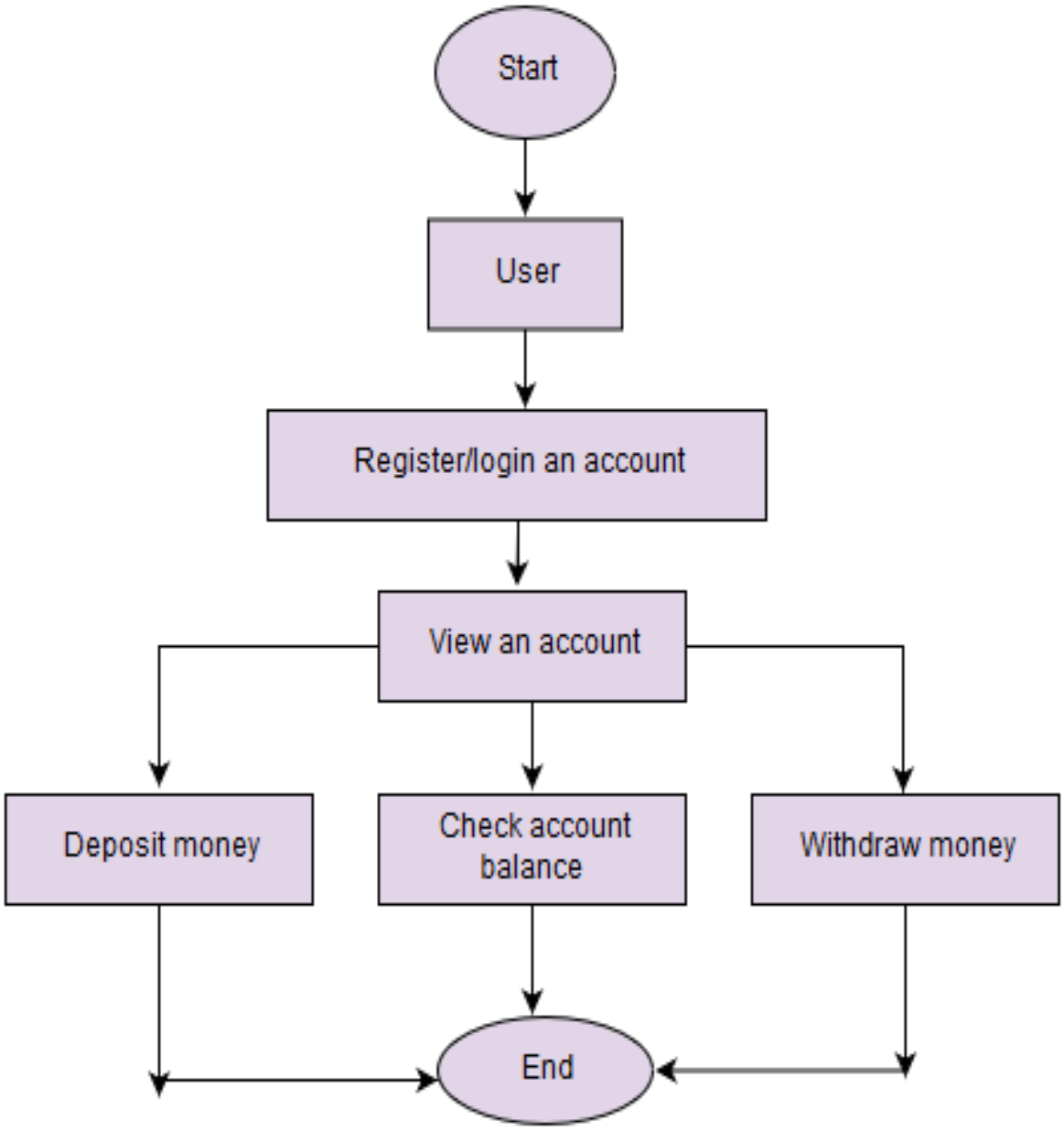


Fig: 3.4 Flow Chart

3.5 Class Diagram

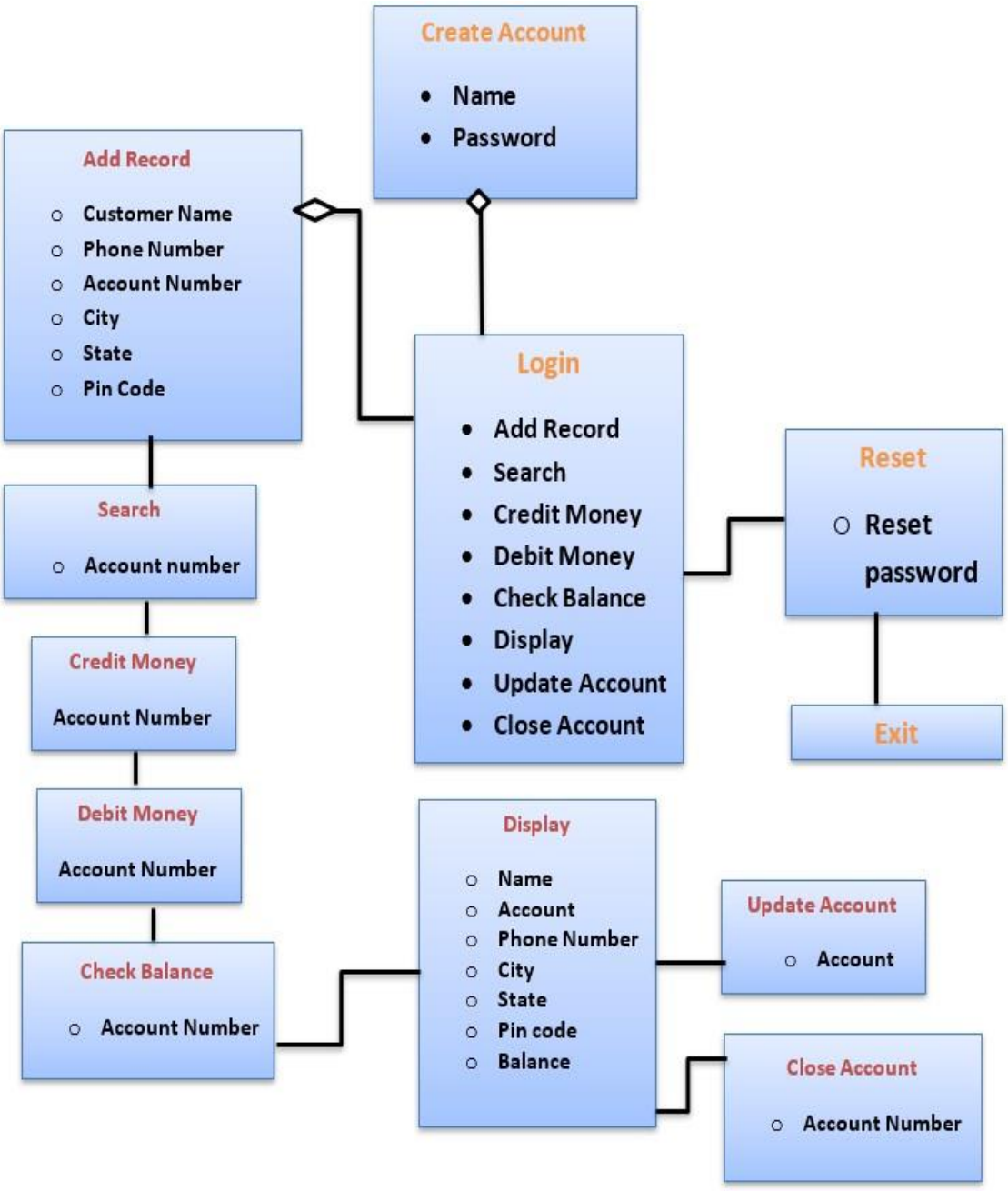


Fig: 3.5 Class Diagram

Chapter 4

Implementation

4. IMPLEMENTATION

4.1 Tools:

Dev -C++ is a free, open-source IDE for C and C++ programming. It was originally developed by Bloodshed Software, but later versions (like Orwell Dev-C++ and Embarcadero Dev-C++) have improved and modernized it.

Key Features:

1. IDE+ Compiler: Bundles the TDM-GCC compiler (a variant of GCC for Windows).
2. Syntax highlighting
3. Code completion
4. Integrated debugger (GDB)
5. Project management tools
6. Customizable interface
7. Light weight and easy to use

4.2 Latest Versions:

There are a few versions of Dev-C++ floating around:

1. Bloodshed Dev-C++ – Very outdated.
2. Orwell Dev-C++ – More updated and stable, popular among Windows users.
3. Embarcadero Dev-C++ – The most recent and actively maintained version.

Chapter 5

Experimental Results and It's Analysis

5. EXPERIMENTAL RESULT AND IT'S ANALYSIS

The Bank Management System project consists of a class named account that represents a bank account. It includes functions for creating an account, displaying account details, modifying account information, depositing, and withdrawing money.

5.1 Create Account:

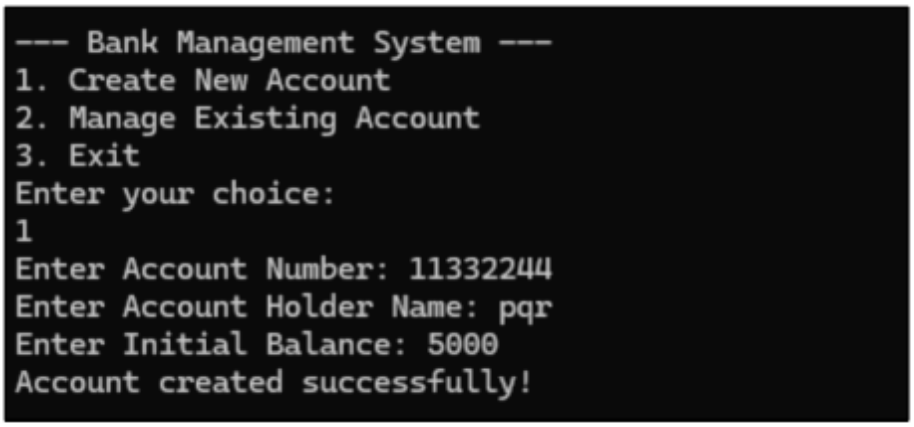


Fig: 5.1 Create Account

5.2 Manage Existing Account:

5.2.1 Desposit:

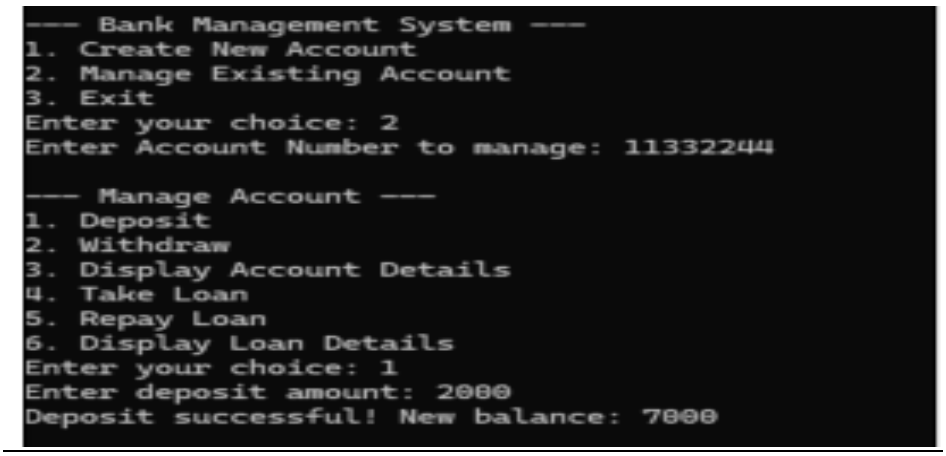


Fig: 5.2 Deposit

5.2.2 Withdraw:

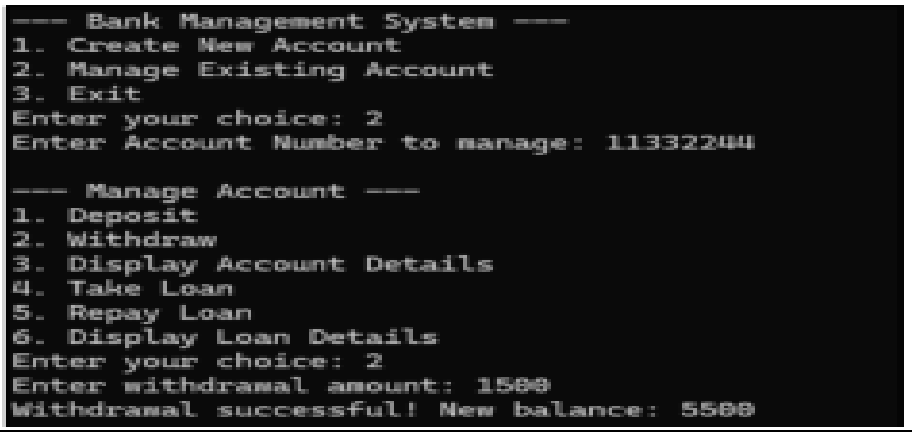


Fig: 5.2.2 Withdraw

5.2.3 Display Account Details:

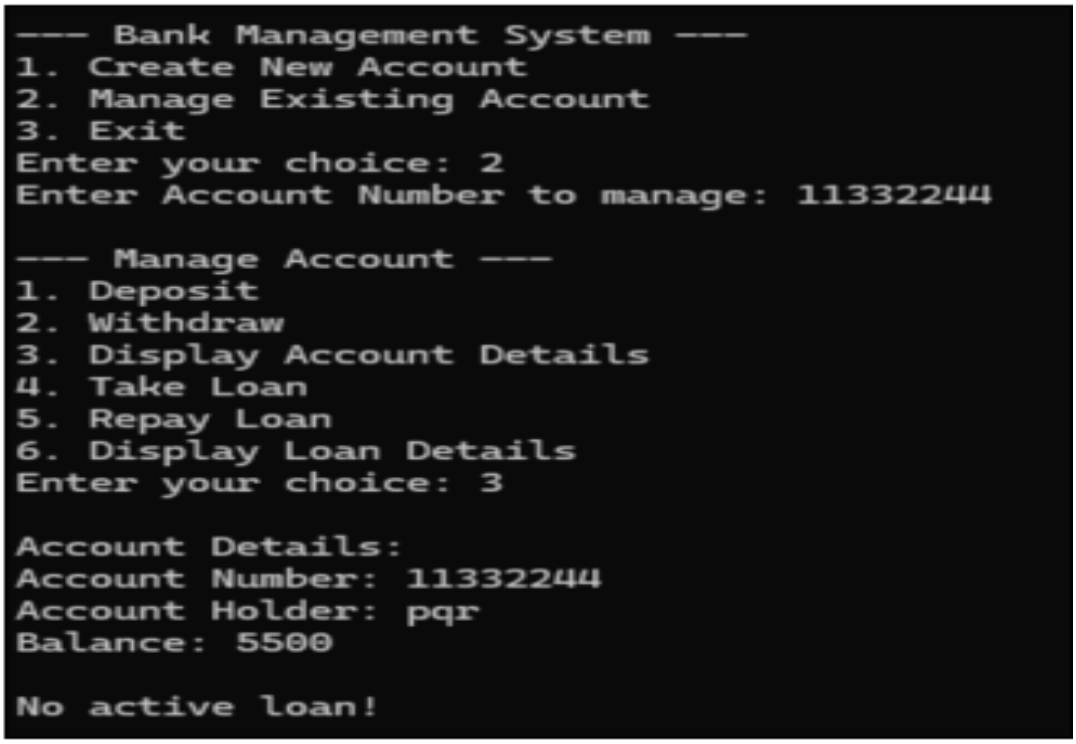


Fig: 5.2.3 Display Account Details

5.2.4 Take Loan:

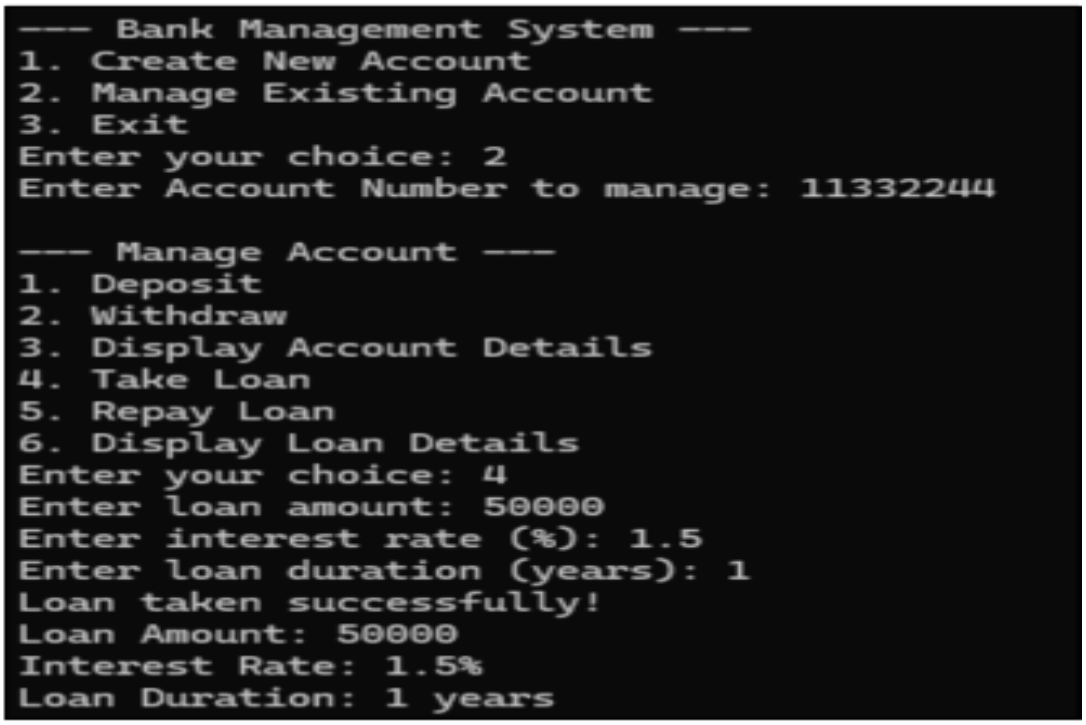


Fig: 5.2.4 Take Loan

5.2.5 Repay Loan:

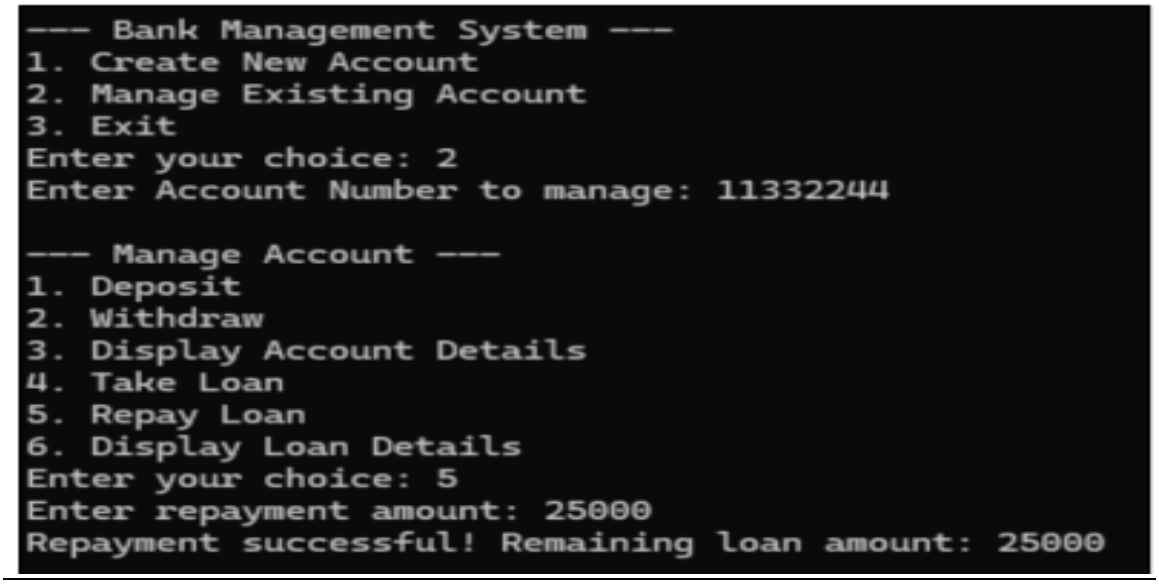


Fig: 5.2.5 Repay Loan

5.2.6 Display Loan Details:

```
--- Bank Management System ---
1. Create New Account
2. Manage Existing Account
3. Exit
Enter your choice: 2
Enter Account Number to manage: 11332244

--- Manage Account ---
1. Deposit
2. Withdraw
3. Display Account Details
4. Take Loan
5. Repay Loan
6. Display Loan Details
Enter your choice: 6

Loan Details:
Loan Amount: 25000
Interest Rate: 1.5%
Loan Duration: 1 years
```

Fig: 5.2.6 Display Loan Details

5.3 Exit:

```
--- Bank Management System ---
1. Create New Account
2. Manage Existing Account
3. Exit
Enter your choice: 3
Thank you for using the Bank Management System!

--------------------------------
Process exited after 235.3 seconds with return value 0
Press any key to continue . . .
```

Fig: 5.3 Exit

Chapter 6

Conclusion

6. CONCLUSION

Conclusion:

The Bank management system with C++ is a basic replica of logic for different operations that can be performed in a bank for opening accounts and managing the same by the customer. This project aids in learning how real-world systems can be modeled and implemented in C++ and demonstrates practical usage of file handling for data persistence.

Chapter 7

References

7. References

7.1 Paper References:

[1] Bank Management and Financial Services by Peter S. Rose and Sylvia C. Hudgins

[2] Bank Management and Regulation: A Book of Readings edited by Stephen A. Rhoades

[3] Bank Management by Timothy W. Koch and S. Scott MacDonald

[4] Banking and Financial Institutions: A Guide for Directors, Investors, and Borrowers by Benton E. Gup

7.2 Web References:

[1] **https://www.bing.com**

[2] **bank-management-system·GitHubTopics·GitHub**

[3] **Menu-DrivenProgramforBankManagementSystem -GeeksforGeeks**

[4] **https://myprojectideas.com/bank-management-system-with-c/**

[5] **https://t4tutorials.com**

[6] **https://copyassignment.com**

7.3 Book References:

[1] "Banking Theory and Practice" Author: K.C. Shekhar & Lekshmy Shekhar

[2] "Management of Banking and Financial Services" Author: Justin Paul & Padmalatha Suresh

[3] "Principles and Practices of Banking" Author: Indian Institute of Banking & Finance (IIBF)

[4] "Banking: An Introduction" Author: H.R. Machiraju

[5] "Information Technology and Its Applications in Business" Author: S.A. Kelkar